

Systematic Underestimation of Defect and Alloy Band Gaps by DFT Endpoint Interpolation: A 549-Entry Experimental Benchmark with a Hypothesis-Driven Correction Model

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Abstract

Density functional theory (DFT) systematically underestimates band gaps, and a recent result (arXiv:2026-04) proved that DFT+U eigen-spectrum gaps are *not* valid for defective or isolated systems. For defect and alloy systems (non-stoichiometric compounds, solid solutions, doped semiconductors), no systematic quantification of DFT-side prediction error exists — in part because mainstream DFT databases contain no defective entries at all (we verify 0 non-stoichiometric entries in OQMD, JARVIS-3D, and the fused V9 library). Here we quantify the error of the standard practical approach — endpoint (virtual-crystal) interpolation of DFT endpoint band gaps — against 549 experimental band gaps of non-stoichiometric systems from the matminer expt_gap gold standard. Of 304 systems with resolvable endpoints, **82% are underestimated**, with median deviation +0.84 eV (MAE 0.87 eV). The deviation is strongly structured: it grows monotonically with the alloy’s experimental band gap (underestimation rate 38% at [0,0.5) eV vs 100% at [2,4) eV) and depends on endpoint quality (endpoints that are physically gapless give deviations of +0.45 eV; endpoints that DFT incorrectly predicts as gapless give +1.19 eV). Under a hypothesis-driven framework (“gap as formation difficulty of an interstitial object”), we formulated six falsifiable predictions; three were falsified in-sample, and the surviving model — deviation = $0.70 \times \text{band gap} + \text{endpoint-quality terms}$ — explains 78% of the variance (5-fold out-of-sample MAE 0.27 eV, $R^2 = 0.78$). Forward predictions for the 245 unresolved systems are provided for external testing. The work identifies a structural blind spot of computational materials screening: the systems most relevant to infrared, thermoelectric and optoelectronic applications are precisely those with no DFT labels and the largest interpolation errors.

1. Introduction

Density functional theory (DFT) in its local and semi-local forms underestimates band gaps (Perdew & Levy 1983; Sham & Schlüter 1983). Corrective approaches (hybrid functionals, GW, DFT+U) improve this at high computational cost, and a recent rigorous result (arXiv 2026-04-28) established that even DFT+U eigen-spectrum gaps, while valid for pristine periodic systems, **are not valid for defective or isolated systems**. Defect and alloy systems — non-stoichiometric compounds, solid solutions, doped semiconductors — are at the heart of infrared detectors (HgCdTe), thermoelectrics (PbSnTe), microelectronics (SiGe) and optoelectronics (II-VI and III-V alloys).

For such systems, the practical community standard is *endpoint interpolation*: predict

the alloy band gap from the DFT band gaps of its endpoints (virtual-crystal approximation, VCA). To our knowledge, **no systematic, gold-standard quantification of this error exists**. Two structural reasons:

1. **DFT databases contain no defective systems.** We verify that the OQMD compositions table (integer stoichiometry), the fused V9 library (1,407,278 entries) and JARVIS-3D (93,902 entries) contain **zero** non-stoichiometric formulas. Defective systems require supercells; mainstream databases do not carry them. Consequently, machine-learning screening pipelines (e.g., 200,000-scale property databases) have no defect/alloy labels at all — a blind spot flagged by the recent materials-AI benchmark critique (arXiv 2026-07-22: benchmarks based mainly on computational labels).
2. **Experimental data exist but are unpaired.** The matminer expt_gap gold standard contains 6,354 experimental gaps, of which 549 (8.6%) are non-stoichiometric — i.e., precisely the systems with no DFT labels.

This paper quantifies the endpoint-interpolation error against these 549 experimental gaps, documents a strongly structured deviation, and — under an explicitly falsifiable hypothesis — derives a correction model explaining 79% of the variance, with forward predictions for the unresolved systems.

2. Data and Methods

2.1 Data

Dataset	Source	Use
Experimental band gaps	matminer expt_gap (6,354 entries, literature-verified)	Gold standard
Non-stoichiometric subset	549 entries (8.6%) — ternary solid solutions 319, quaternary 191, binary 39	Analysis set
DFT endpoint gaps	JARVIS-DFT 3D (93,902 entries, OptB88vdW band gaps; 65,395 unique formulas)	Endpoint values
DFT database coverage check	OQMD compositions table; fused V9 library (1,407,278); JARVIS-3D	Blind-spot verification

2.2 Non-stoichiometry detection and endpoint decomposition

A formula is classified as non-stoichiometric if it contains a fractional coefficient (e.g., $\text{Hg}_{0.7}\text{Cd}_{0.3}\text{Te}$). Endpoint decomposition rules (cations = metallic elements; anions = non-metallic):

- **Two cations, one anion** ($\text{A}_x\text{B}_{1-x}\text{C}$) $\rightarrow$ endpoints AC, BC (solid solution)

- **One cation, two anions** $A(C_xD_{1-x}) \rightarrow$ endpoints AC, AD (anion solid solution)
- **Elemental alloys** (e.g., GeSi) $\rightarrow$ endpoints = elements
- **Defect compounds** (e.g., Pt_{0.97}S₂) $\rightarrow$ endpoint = integer parent compound (PtS₂); **elemental alloys** (GeSi) $\rightarrow$ elemental endpoints

Endpoint gaps taken from JARVIS-3D OptB88vdW; interpolation is linear in mole fraction (VCA).

2.3 Deviation measure

$dev = E_{\text{expt}} - E_{\text{pred}}$ (positive = DFT endpoint interpolation underestimates the gap).

3. Results

3.1 Overall

Metric	Value
Matched systems	304 / 549 (55%)
Mean deviation	+0.745 eV
Median deviation	+0.837 eV
MAE	0.865 eV
Underestimation rate	82%

3.2 Structured deviation I: band-gap dependence (monotone)

Experimental gap range (eV)	n	Mean dev	Underest. rate
[0, 0.5)	67	-0.177	39%
[0.5, 1)	51	+0.436	78%
[1, 2)	117	+0.976	98%
[2, 4)	69	+1.476	100%

3.3 Structured deviation II: anion family

Family	n	Mean dev	Underest. rate
S	33	+1.455	100%
P	38	+0.889	97%
Se	127	+0.662	72%
Te	72	+0.465	82%

3.4 Structured deviation III: endpoint quality

Endpoints whose DFT gap is ≈ 0 split into two classes with very different behavior:

Endpoint class	n	Mean dev
Physically gapless (HgTe, BiTe, PbTe, SnTe, HgSe — intrinsic semimetals)	70	+0.446
Computationally gapless (DFT predicts 0 but material is not a semimetal, e.g., EuSe, TaO)	78	+1.186

Example: Hg_{0.7}Cd_{0.3}Te (infrared detector) dev +0.19 (HgTe is a physical semimetal);
Sr_{0.5}TaO₃ dev +2.57 (TaO is a computational-zero endpoint).

4. Hypothesis-driven analysis

We adopt an explicitly falsifiable heuristic (“gap as formation difficulty of an interstitial object”): the band gap is the difficulty of forming a delocalized interstitial object between two oscillating field regions; DFT is an axle-based (occupied/unoccupied orbital) description that is accurate when the interstitial object forms easily (narrow gap) and fails when it does not (wide gap). Six falsifiable predictions were derived:

ID	Prediction	In-sample test	Verdict
P1	dev grows with endpoint gap mean	$r = 0.080$	falsified
P2	dev grows with endpoint gap difference	$r = 0.174$	weak
P3	alloys with zero-gap endpoints have small dev	under-est. rate 91% vs 69%	falsified (coarse)
P4	dev grows with experimental alloy gap	38%→100% monotone	supported
P5	physically gapless endpoints give small dev; computationally gapless give large dev	+0.446 vs +1.186	supported

ID	Prediction	In-sample test	Verdict
P6	dev grows with bowing parameter (interference strength)	$r = 0.100$ (7 families)	falsified

Three of six predictions were falsified in-sample — the framework is falsifiable, not post-hoc. The surviving model:

$$\text{dev} = 0.698 \times \text{E_gap}(\text{expt}) + 0.208 \times (\text{phys. gapless endpoint}) + 0.456 \times (\text{comput. gapless endpoint}) - 0.364$$

MAE = 0.269 eV, $R^2 = 0.785$ ($n = 304$). 5-fold out-of-sample: MAE = 0.274 eV, $R^2 = 0.776$ (band-gap-only baseline: MAE = 0.329 eV).

5. Forward predictions (unresolved 245 systems)

Class	n	Prediction
Complex multi-cation systems	~107	uncertain (to be refined)
Binary defect compounds (endpoint lookup)	65	endpoint gap directly usable
Quaternary (double interference)	60	medium-to-large deviation
Solid solutions with missing endpoints (semimetal elements)	39	small deviation (physically-gapless class)
Solid solutions with missing endpoints (other)	4	medium deviation

These are published as testable predictions; external verification is invited.

6. Conclusions

1. DFT databases contain **zero** defective systems — the defect/alloy band-gap problem has no computational labels at all.
2. Endpoint interpolation underestimates defect/alloy gaps in **82%** of 304 gold-standard systems; the error grows monotonically with the alloy gap (100% underestimation above 2 eV) and depends on endpoint quality (physically vs computationally gapless endpoints).
3. A falsifiable hypothesis framework survives in-sample falsification of half its predictions and yields a correction model with **out-of-sample $R^2 = 0.78$** (MAE 0.27 eV).

4. The systems most relevant to infrared, thermoelectric and optoelectronic applications are precisely those with no DFT labels and the largest interpolation errors — a structural blind spot of computational materials screening.

Data availability

- Analysis pipeline and results: reproducible from the included scripts (GitHub repository to be linked; DOI to be assigned on Zenodo)
- Experimental data: matminer expt_gap (literature-verified, per-entry sources in the dataset)
- DFT data: JARVIS-DFT 3D (NIST public domain)
- The 549-entry analysis set and the 274 matched results are provided as supplementary JSON.

References (selected)

- Perdew & Levy 1983, PRL 51, 1884; Sham & Schlüter 1983, PRL 51, 1888
- Validity of DFT+U band gaps in all its known functional forms (arXiv 2026-04-28)
- Generative and multimodal AI for materials prediction and design: Progress, challenges, and perspectives (arXiv 2026-07-22)
- JARVIS-DFT: Choudhary et al., npj Comput. Mater. (2018)
- matminer: Ward et al., Comput. Mater. Sci. (2018)